COMPx270: Randomised and Advanced Algorithms

Lecture 5: Graph algorithms

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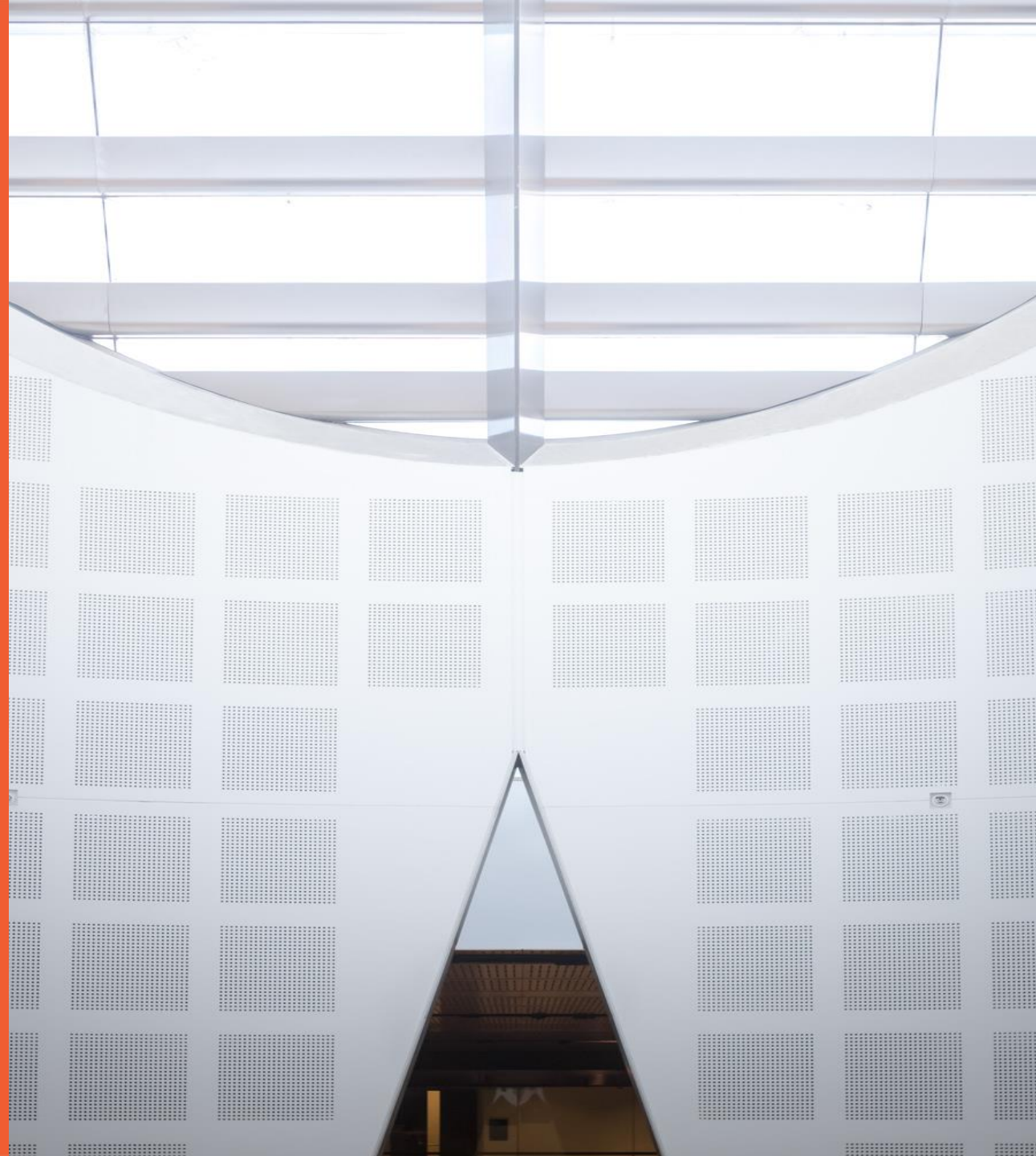
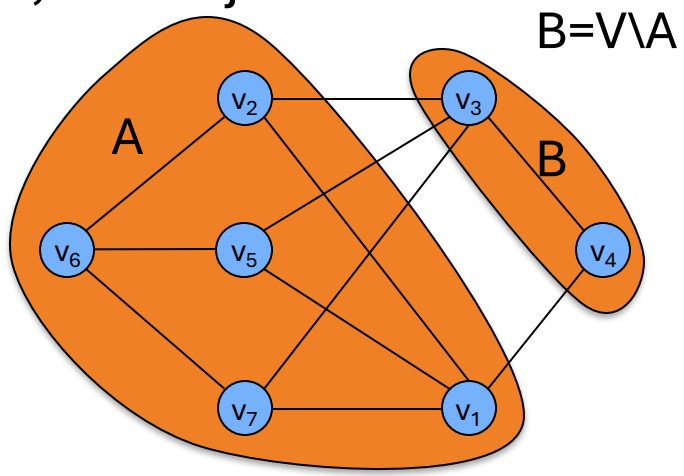


Global Minimum Cut

Input: A connected, undirected graph G = (V, E).

For a set $A \subseteq V$ let $\delta(A) = \{(u,v) \in E : u \in A, v \in V \setminus A\}$.



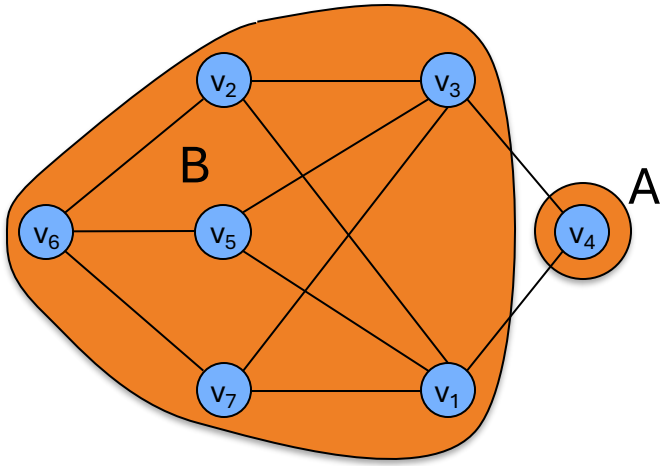
$|\delta(A)| = 4$

Aim: Find a cut (A, B) minimizing $|\delta(A)|$.

Global Minimum Cut

Input: A connected, undirected graph G = (V, E).

For a set $A \subseteq V$ let $\delta(A) = \{(u,v) \in E : u \in A, v \in V \setminus A\}$.



$|\delta(A)| = 2$

Aim: Find a cut (A, B) of minimum cardinality.

Global Minimum Cut

Applications: Partitioning items in a database, identifying clusters of related documents, network reliability, network design, circuit design, TSP solvers.

Network flow solution.

- Replace every edge (u, v) with two directed edges (u, v) and (v, u).
- Pick some vertex s and compute min s-v cut separating s from each other vertex $v \in V$.

Running time: $O((n-1) \cdot \text{MaxFlows})$

Global Minimum Cut

Running time: $O((n-1) \cdot \text{MaxFlows})$ — MaxFlows: ?

F: value of the max flow

Ford-Fulkerson:	$O(mF)$
*:	$O(m \log F)$
Edmonds-Karp:	$O(mn)$
Best (?):	$O(m \log \frac{n^2}{m})$

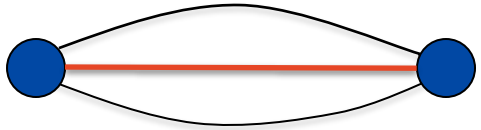
gives $O(n \cdot m \cdot \log \frac{n^2}{m})$ algo for min-cut

but $O(n^3)$ for dense graphs

All these are deterministic

Karger's Contraction Algorithm

Definition. A multigraph is a graph that allows multiple edges between a pair of vertices.



Karger's Contraction Algorithm

Definition. A multigraph is a graph that allows multiple edges between a pair of vertices.

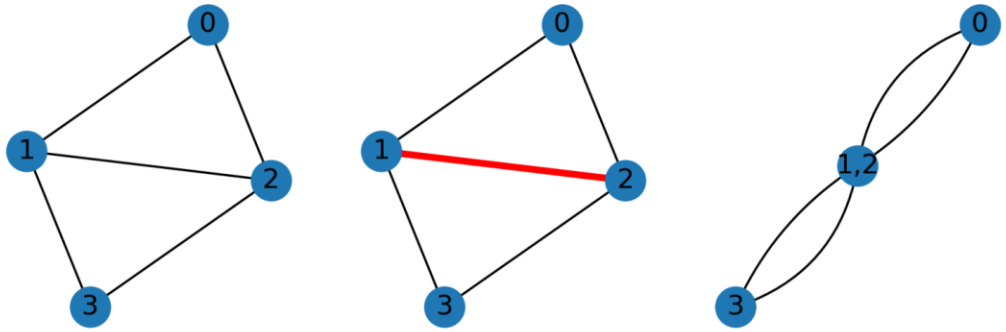
Karger's Contraction Algorithm

Let G=(V,E) be a multigraph (without self-loops).

The contraction of an edge e=(u,v)∈E gives G\e

- Replace u and v by single new super-node w
- Replace all edges (u,x) or (v,x) with an edge (w,x)
- Remove self-loops to w

} can be done in $O(n)$ time



Karger's Contraction Algorithm

(assume graph G is connected, o/w what are we doing here?)

Require: multigraph $G = (V, E)$

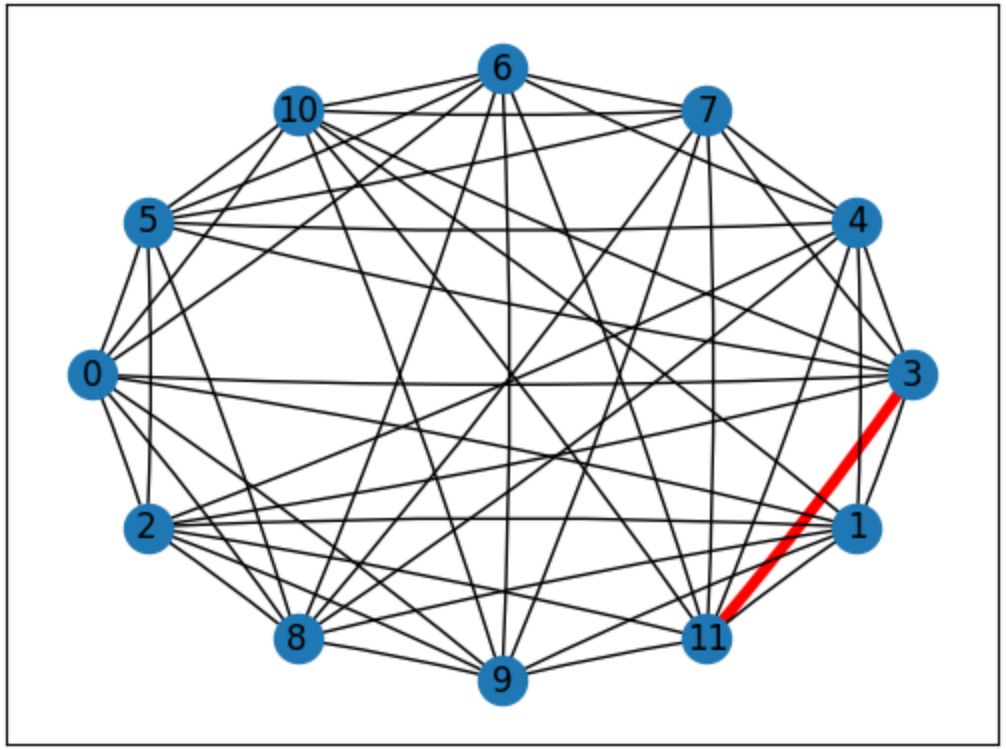
1: **while** $|V| > 2$ **do**
2: $O(n)$ Pick an edge $e \in E$ uniformly at random
3: $O(n)$ Contract it, and let $G \leftarrow G/e$
4: **return** the cut defined by the remaining two vertices.

$O(n^2)$ time

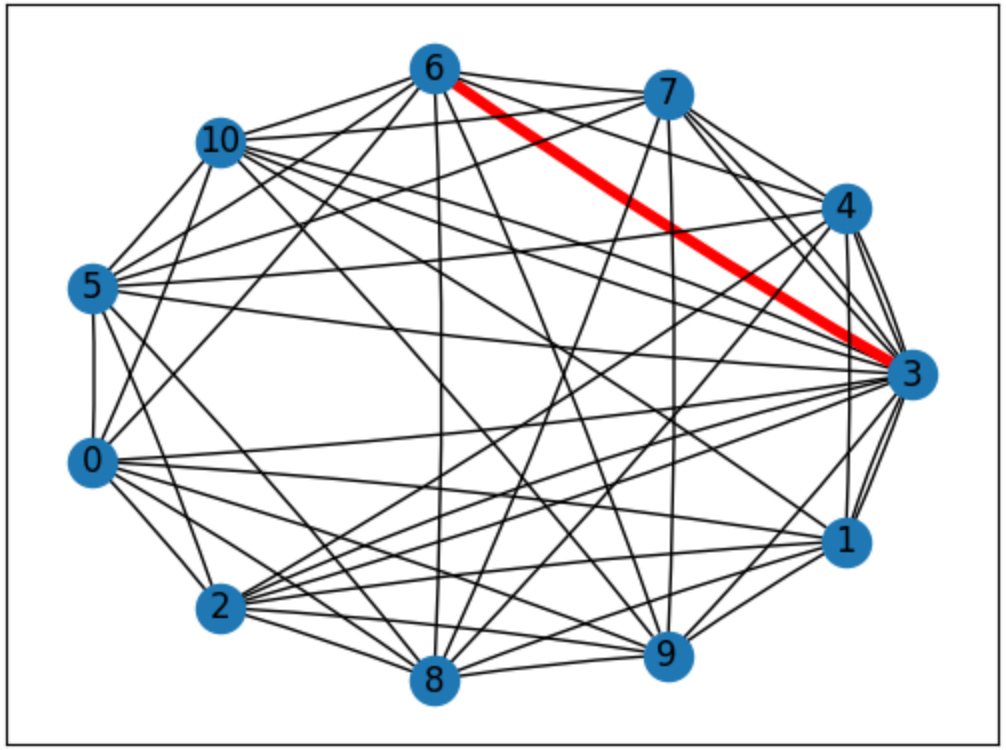
one fewer vertex

$n-2$ iterations

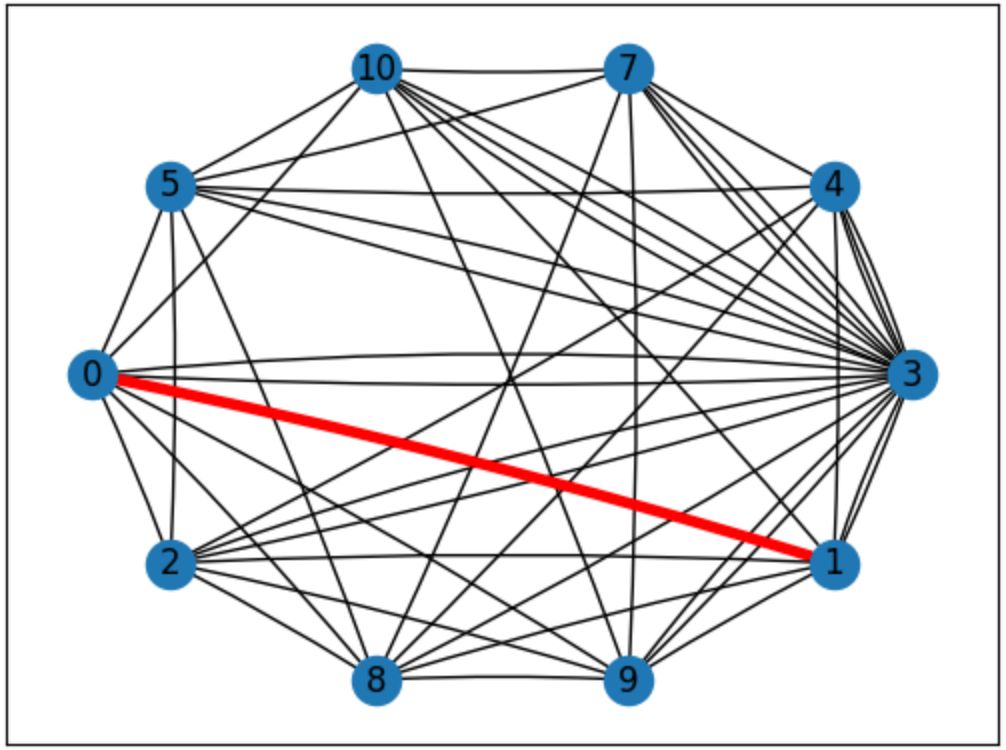
Karger's Contraction algorithm



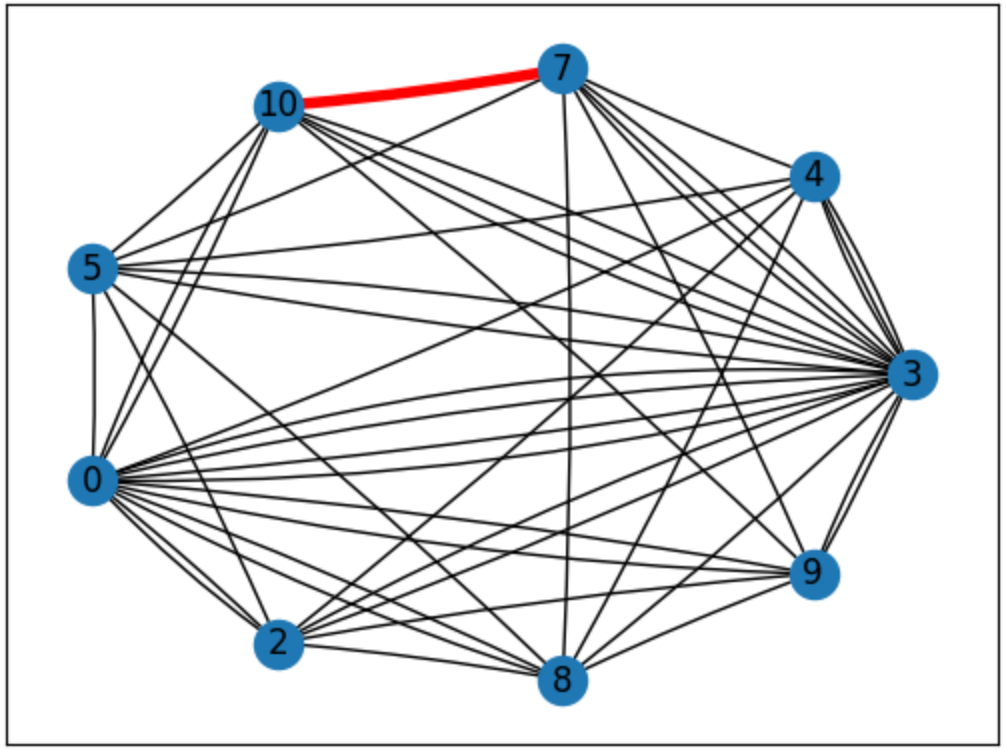
Karger's Contraction algorithm



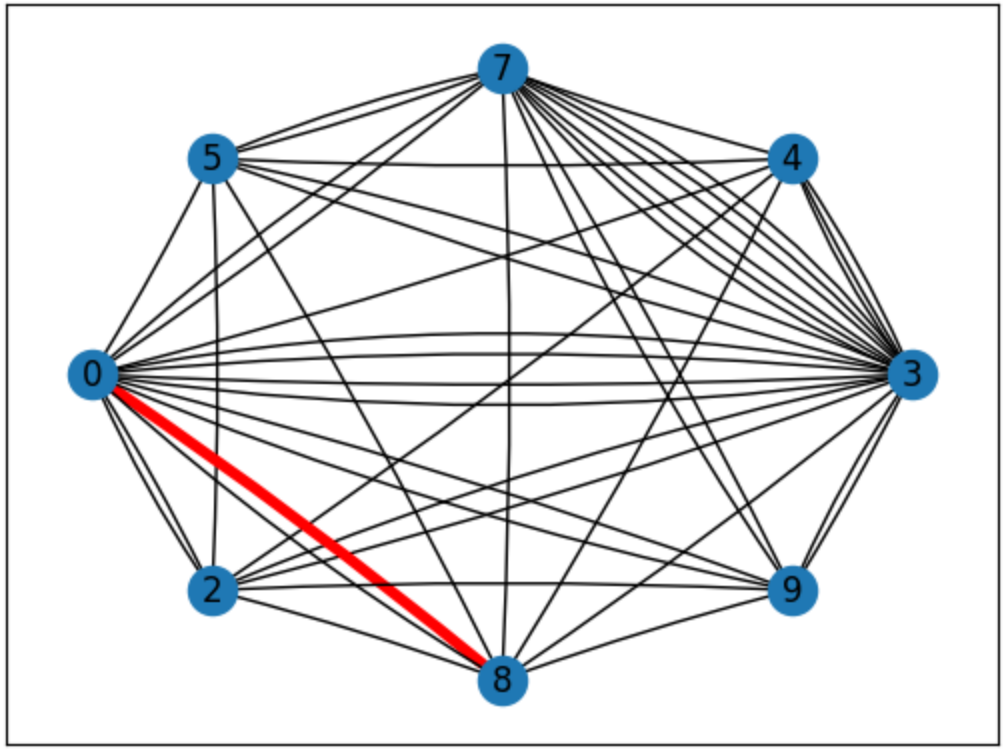
Karger's Contraction algorithm



Karger's Contraction algorithm



Karger's Contraction algorithm



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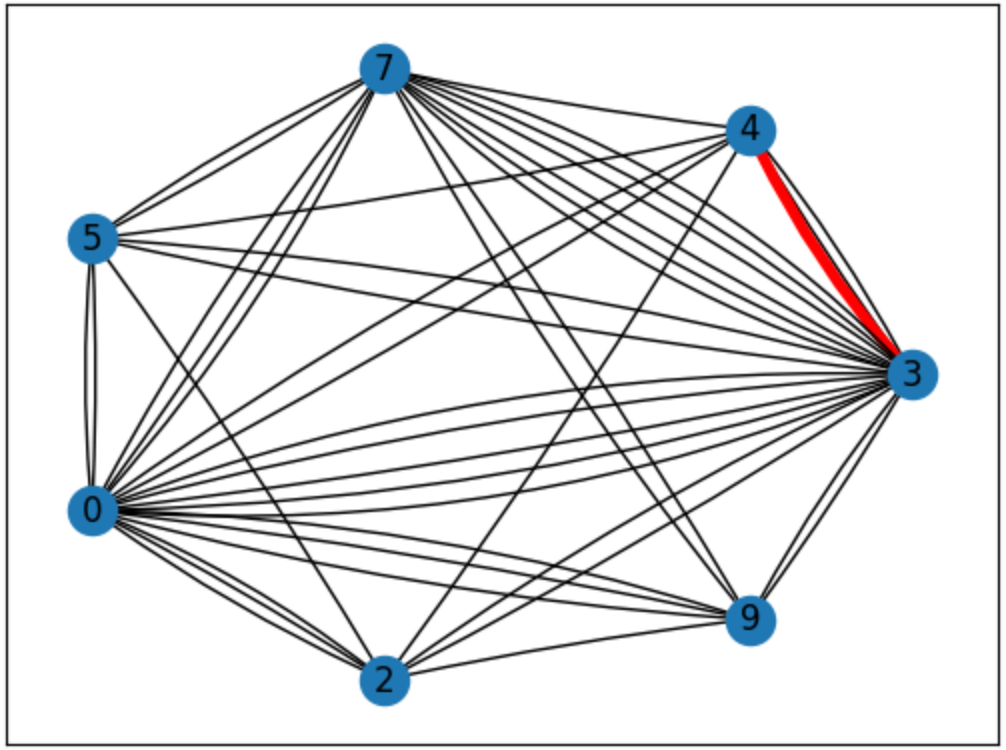

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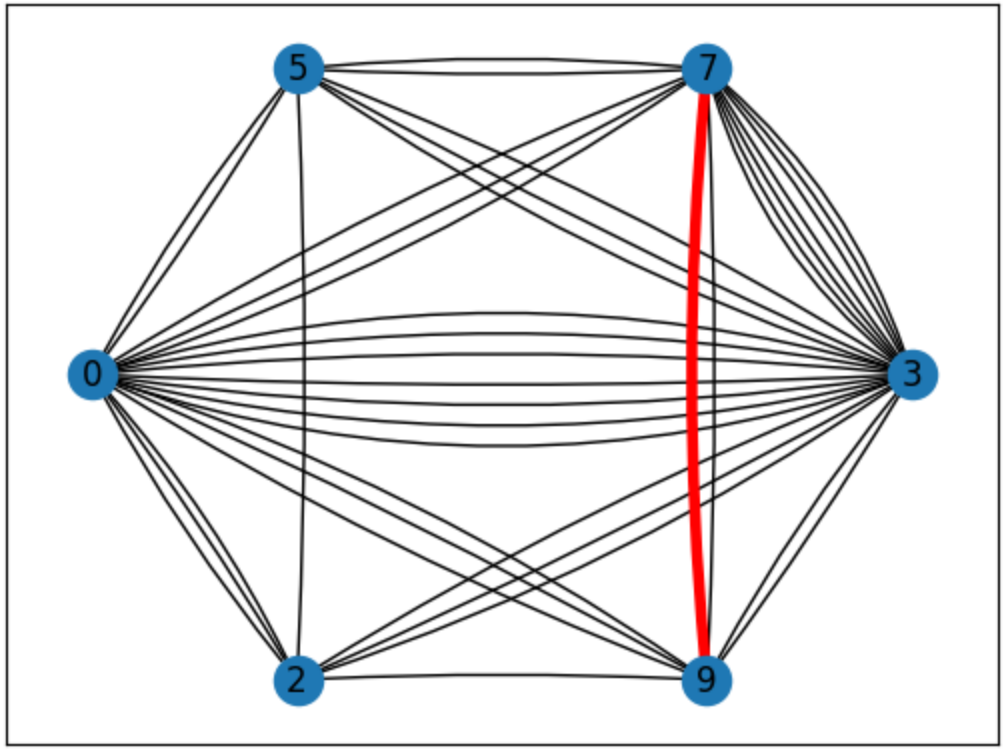

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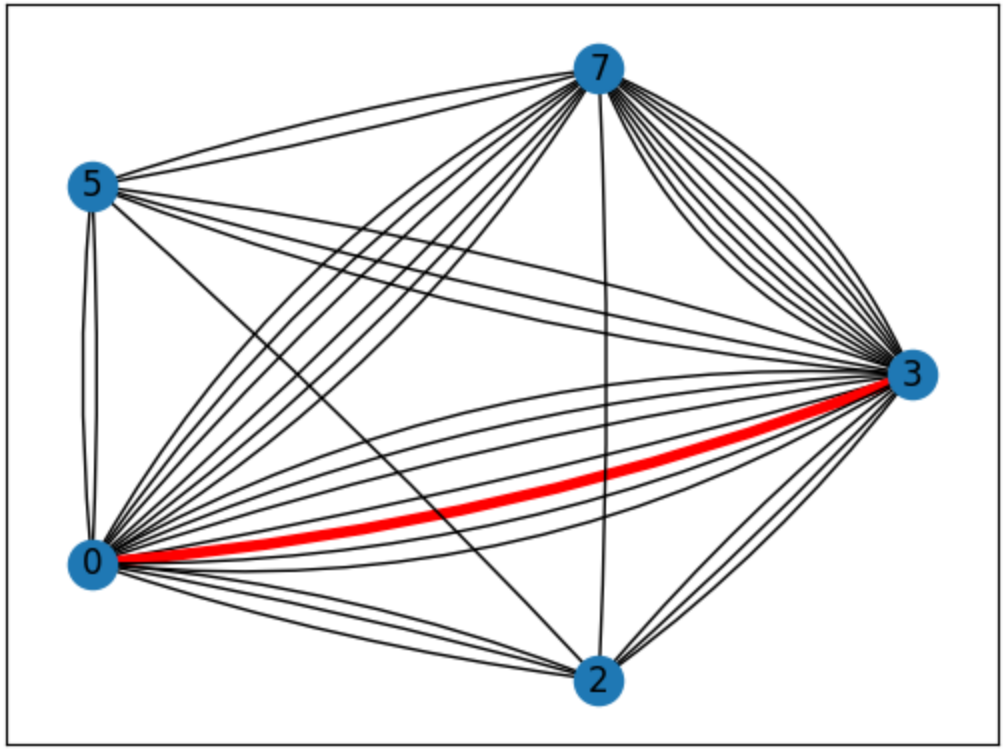
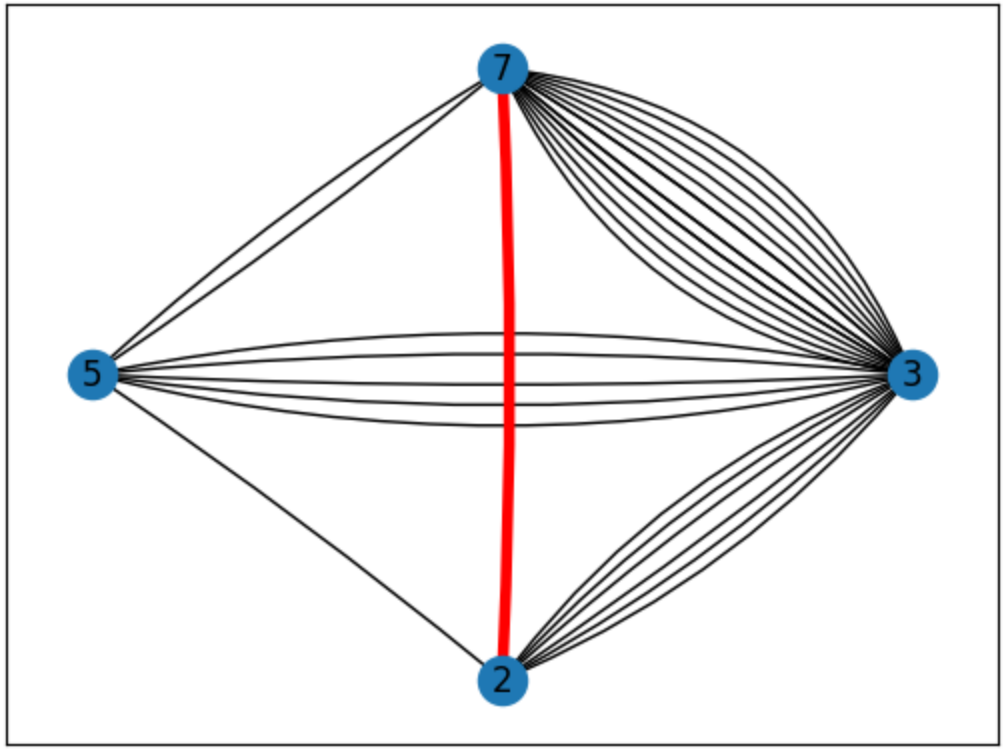
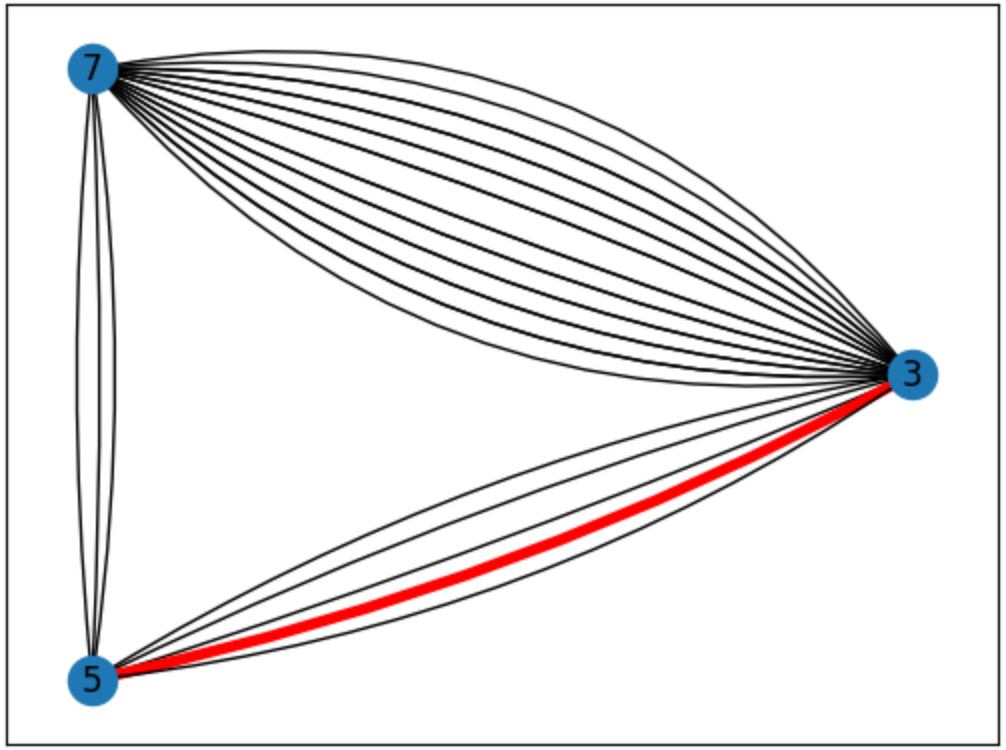


Karger's Contraction algorithm



Karger's Contraction algorithm



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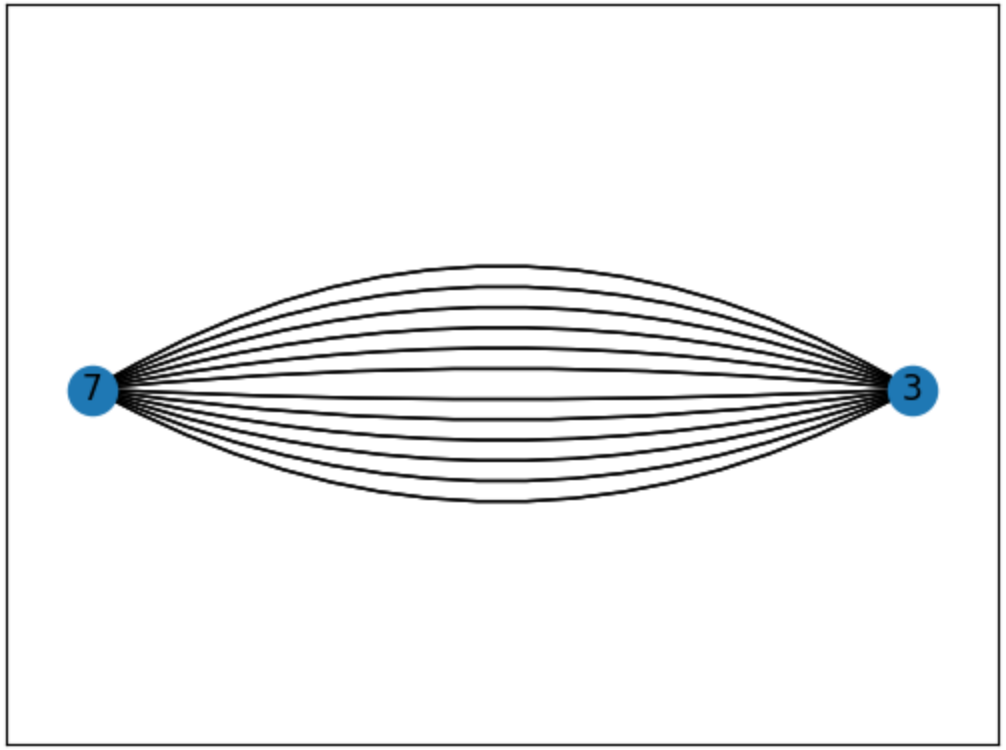
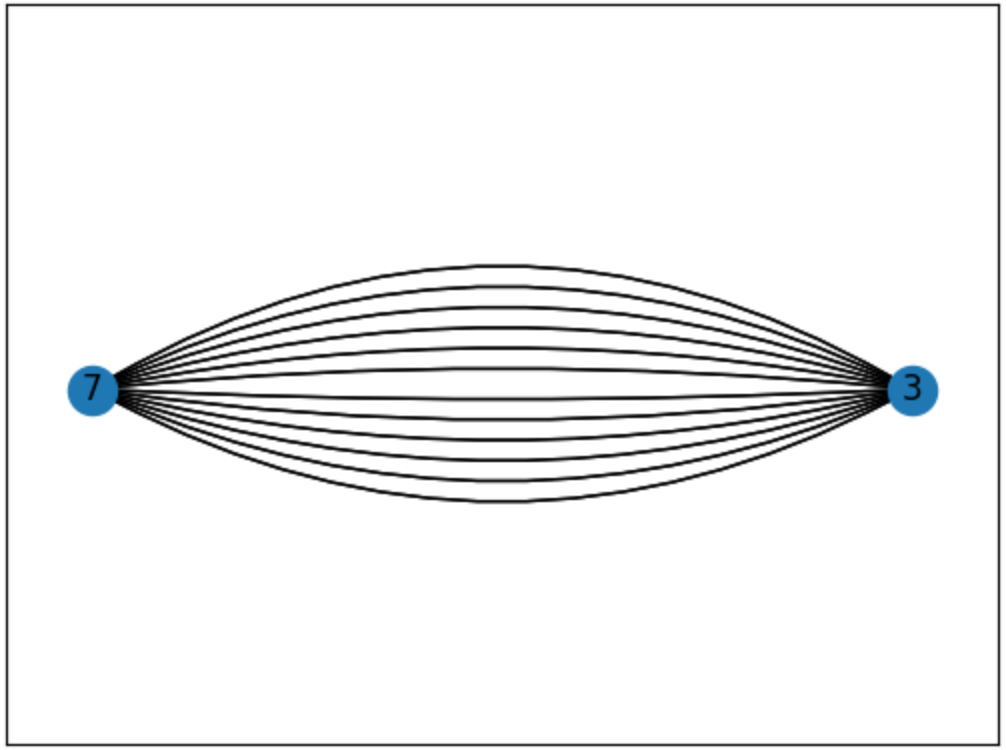


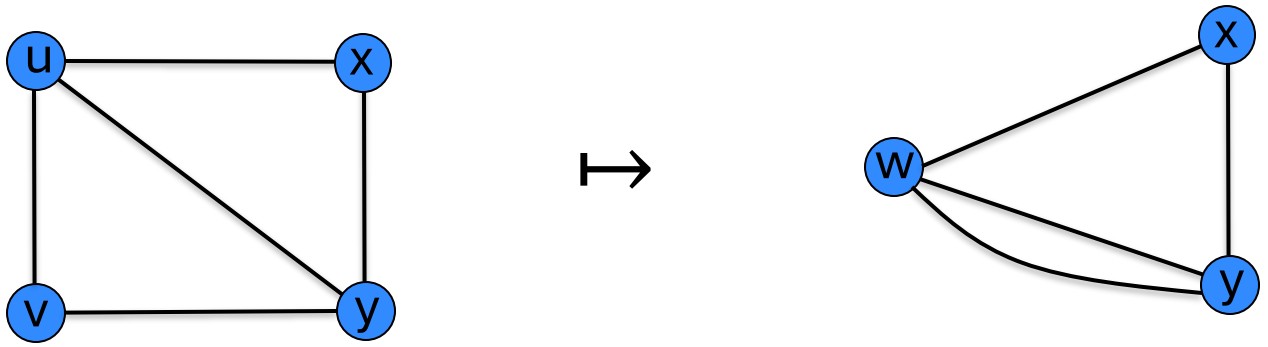
Karger's Contraction algorithm



The End.

Karger's Contraction Algorithm

Observation: An edge (u,v) contraction preserves the cuts (A,B) where u and v are both in A or both in B.



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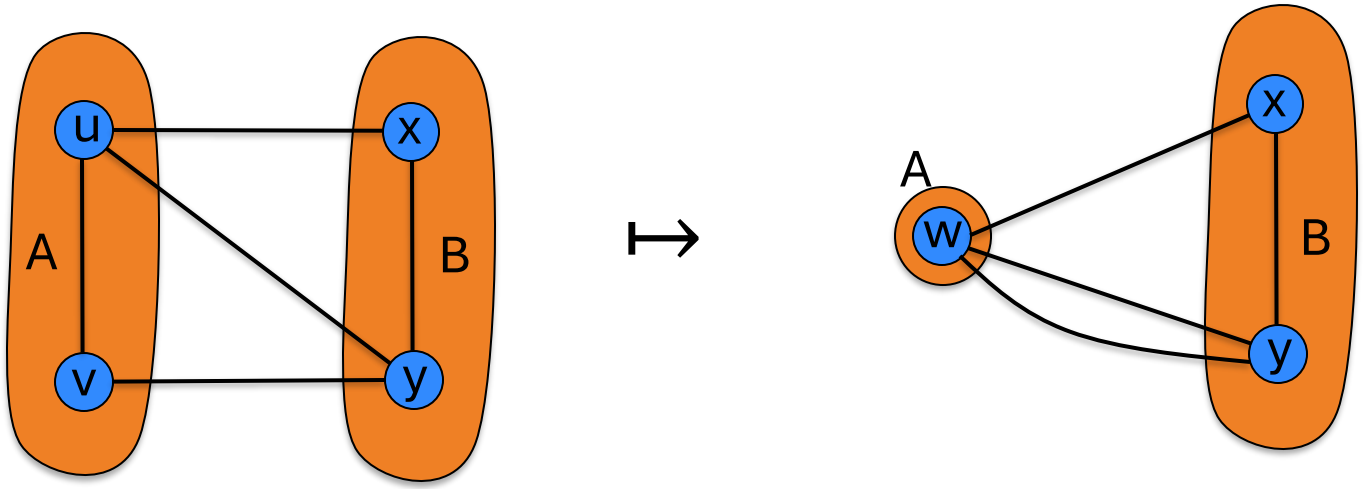


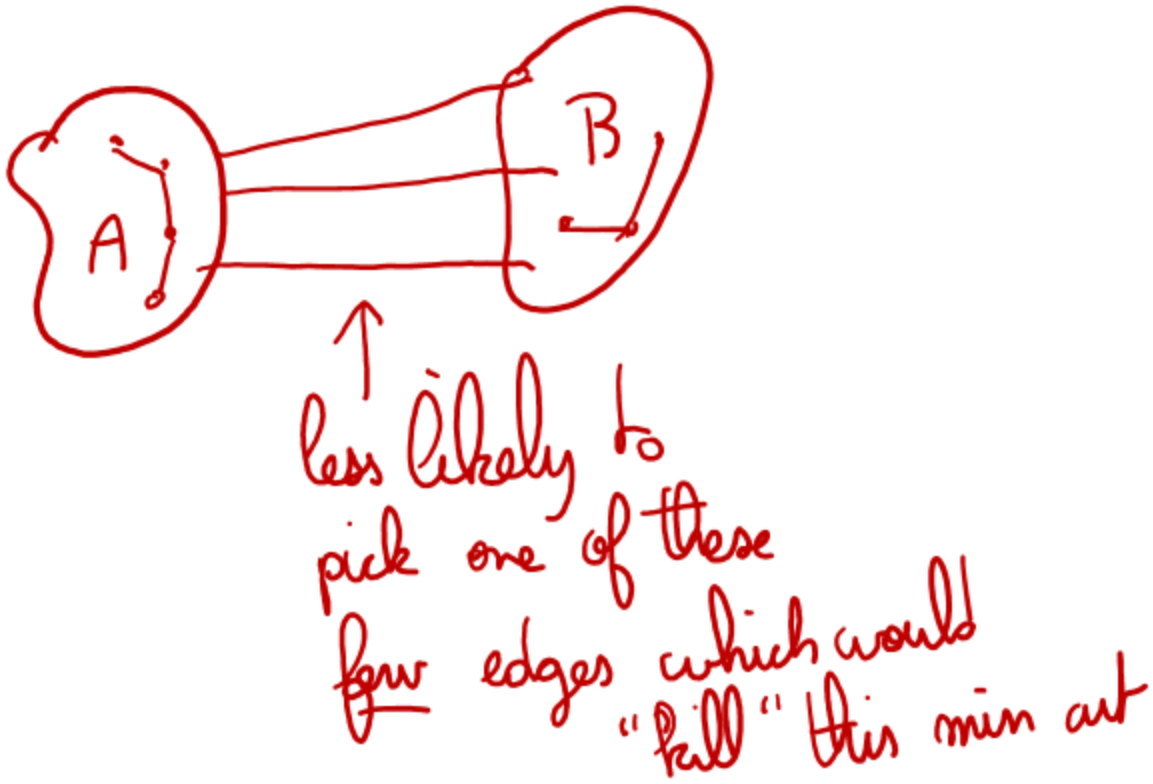
Karger's Contraction Algorithm

Observation: An edge (u,v) contraction preserves the cuts (A,B) where u and v are both in A or both in B.

If $u,v \in A$ then $\delta_G(A) = \delta_{G\setminus e}(A)$.
(with u and v replaced with "uv")

Karger's Contraction Algorithm

Observation: If (A,B) is a minimum cut, then we are less likely to choose an edge (u,v) crossing it!



Karger's Contraction Algorithm

Require: multigraph $G = (V, E)$

1: **while** $|V| > 2$ **do**

2: Pick an edge $e \in E$ uniformly at random

3: Contract it, and let $G \leftarrow G/e$

4: **return** the cut defined by the remaining two vertices.

Claim. This algorithm has a reasonable chance of finding a min cut.

↑

do not want exponentially small!

Key claim

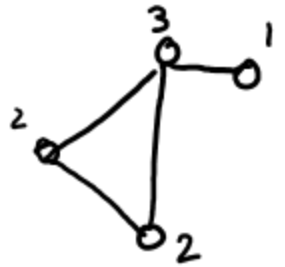
Claim. If C is a min-cut, then the algorithm returns it with probability at least $2/n^2$.

Useful

Handshaking lemma

Let $G=(V,E)$ be a (multi)graph.

Then $$|E| = \frac{1}{2}\sum_{v \in V} \deg v$$



$|E|=4$

$$\sum_{v \in V} \deg v = 2+2+3+1 = 8 = 2|E|$$

Key claim

Claim. If C is a min-cut, then the algorithm returns it with probability at least $2/n^2$.

Proof. Fix a min-cut C, let $k \triangleq |C|$. E_i = event that edge picked at iteration i did not "kill" C

$$\Pr[C \text{ is returned}] = \Pr[C \text{ is never killed}]$$
$$= \Pr[E_1 \cap E_2 \cap \cdots \cap E_{n-2}]$$
$$= \Pr[E_1] \cdot \Pr[E_2 \mid E_1] \cdot \Pr[E_3 \mid E_1 \cap E_2] \cdot \cdots \cdot \Pr[E_{n-2} \mid \bigcap_{i=1}^{n-3} E_i]$$

Let's look at one term:

$$\Pr[E_{i+1} \mid E_1 \cap \cdots \cap E_i]$$

After step i:
- C is still there (hasn't been killed)
- $G \setminus \{e_1, \ldots, e_i\} \triangleq G_i = (V_i, E_i)$
- $|V_i| = n - i$
- $|E_i| =$??

$$\Pr\Big[\underbrace{C \text{ is killed at step } i+1}_{E_{i+1}^c} \;\Big|\; \underbrace{C \text{ not killed so far}}_{\bigcap_{\ell \le i} E_\ell}\Big] = \frac{|C|}{|E_i|} = \frac{k}{|E_i|} \le \frac{2}{n-i}$$

$$|E_i| \underset{HSL}{=} \frac{1}{2}\sum_{v \in V_i} \deg v \ge \frac{1}{2}(n-i)k$$

↑ if $\exists\, v$ w/ $\deg v < k$ in E_i, corresponds to a cut in G (original graph) better than C!

and so

$$\Pr\Big[E_{i+1} \,\Big|\, \bigcap_{\ell \le i} E_i\Big] \overset{\text{A}}{\ge} 1 - \frac{2}{n-i}$$

$$\Pr[C \text{ survives}] = \prod_{i=0}^{n-3} \Pr[E_{i+1} \mid E_1 \cap \dots \cap E_i] \overset{\text{A}}{\ge} \prod_{i=0}^{n-3}\left(1 - \frac{2}{n-i}\right) = \prod_{i=0}^{n-3} \frac{n-i-2}{n-i}$$

$$= \prod_{j=3}^{n} \frac{j-2}{j} = \frac{1 \cdot 2 \cdot 3 \cdot 4 \cdot \dots \cdot (n-2)}{3 \cdot 4 \cdot \dots \cdot (n-2)(n-1)n} = \frac{2}{(n-1)n} = \frac{1}{\binom{n}{2}}$$

$$\ge \frac{2}{n^2} \qquad \square$$

Amplification

To amplify the probability of success, run the contraction algorithm many times.

Require: multigraph $G = (V, E)$, integer T

1: **for** $1 \leq t \leq T$ **do** ▷ Use fresh (independent) random bits for each

2: Run Algorithm on G, let C_t be the output

3: **return** the smallest cut among all cuts $C_1, \ldots, C_T$ obtained

Amplification

To amplify the probability of success, run the contraction algorithm many times.

Claim. If we repeat the contraction algorithm $r\binom{n}{2}$ times with independent random choices, the probability that all runs fail is at most $(1/e)^r$.

$$\Pr[\text{no min cut is returned in } r\tbinom{n}{2} \text{ runs}] \le \left(1 - \frac{1}{\binom{n}{2}}\right)^{r\binom{n}{2}} = \left(\left(1 - \frac{1}{\binom{n}{2}}\right)^{\binom{n}{2}}\right)^{r}$$

$$\le \left(\left(e^{-\frac{1}{\binom{n}{2}}}\right)^{\binom{n}{2}}\right)^{r} = e^{-r}$$

$1 - x \le e^{-x}$

$1 + x \le e^{x}$ ★

run $T = \underbrace{\Theta(n^2)}_{r = \Theta(1) \text{ in the thing above}}$ times to get success probability $1 - \frac{1}{100} = 99\%$

Karger's Contraction Algorithm

Require: multigraph $G = (V, E)$

1: **while** $|V| > 2$ **do**
2: Pick an edge $e \in E$ uniformly at random
3: Contract it, and let $G \leftarrow G/e$
4: **return** the cut defined by the remaining two vertices.

$n-2$ iterations

$O(n)$ time (carefully)

Running time?

$O(n^2)$ for success proba $\frac{2}{n^2}$

$O(Tn^2) = O(n^4)$ for success proba 99% :(

Karger's Contraction Algorithm

Require: multigraph $G = (V, E)$

1: **while** $|V| > 2$ **do**

2: Pick an edge $e \in E$ uniformly at random

3: Contract it, and let $G \leftarrow G/e$

4: **return** the cut defined by the remaining two vertices.

Running time?

The algorithm is iterated $O(n^2 \log(1/\delta))$ times... total running time $O(n^4 \log(1/\delta))$.

Karger's Contraction Algorithm

Can we do better?

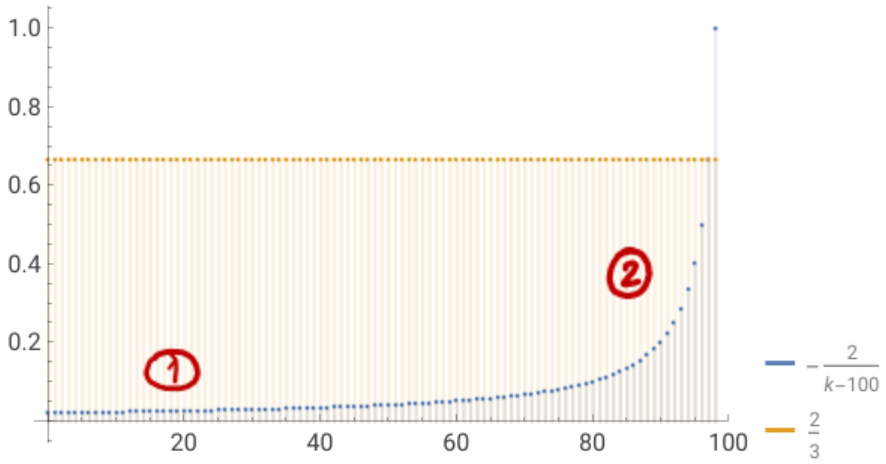
Karger's Contraction Algorithm

Can we do better?

Fix any min cut C.

$$\Pr[\,C \text{ is killed at step } i+1 \mid C \text{ survived until then}\,] \leq \frac{2}{n-i}$$

↑ kind of tight

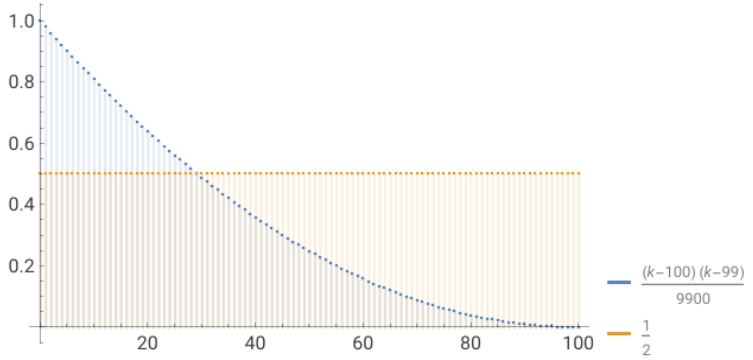


- At first, kill C
  - ① w/ very low proba : good
- At the end very likely to kill C : bad
  - ②

$$\Pr[\,C \text{ is killed after } n-s \text{ steps}\,] \leq \frac{s(s-1)}{n(n-1)} \approx \frac{s^2}{n^2}$$

Improved algorithm

Improvement. [Karger-Stein 1996]



- Early iterations are less risky than later ones: probability of contracting an edge in min cut hits 50% when $n/\sqrt{2}$ nodes remain.
- Run contraction algorithm until $n/\sqrt{2}$ nodes remain.
- Run contraction algorithm twice on resulting graph, and return best of two cuts.

Improved algorithm

```
1: procedure ModifiedKarger(G = (V, E), s)
2:     while |V| > s do
3:         Pick an edge e ∈ E uniformly at random
4:         Contract it, and let G ← G/e
5:     return G
6: procedure KargerStein(G = (V, E))
7:     if |V| ≤ 6 then
8:         return a minimum cut                ▷ Brute-force computation
9:     Set s ← ⌈n/√2 + 1⌉
10:    ▷ Contraction
11:        G_1 ← ModifiedKarger(G, s)
12:        G_2 ← ModifiedKarger(G, s)
13:    ▷ Recursion
14:        C_1 ← KargerStein(G_1)
15:        C_2 ← KargerStein(G_2)
16:    return the smallest cut among C_1, C_2
```

Improved algorithm: Karger-Stein

$O(|V|^2)$

```
1: procedure ModifiedKarger(G = (V, E), s)
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11:        G1 ← ModifiedKarger(G, s)
12:        G2 ← ModifiedKarger(G, s)
13:    ▷ Recursion
14:        C1 ← KargerStein(G1)
15:        C2 ← KargerStein(G2)
16:    return the smallest cut among C1, C2
```

] $O(1)$

2 calls to $O(|V|^2)$

2 recursive calls

Running time?

$T(n) = 2T(\frac{n}{\sqrt{2}}) + O(n^2)$

recursive calls

modified Karger

$\rightsquigarrow T(n) = O(n^2 \log n)$

Improved algorithm: Karger-Stein

Running time?

```
1: procedure ModifiedKarger(G = (V, E), s)
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Improved algorithm: Karger-Stein

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Success probability?

Much better!

$\Omega\left(\frac{1}{\log n}\right)$

Improved algorithm: Karger-Stein

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16:    return the smallest cut among C_1, C_2
```

Success probability?

$\underbrace{\Pr[\text{success}]}_{p(n)} = \Pr[C_1 \text{ or } C_2 \text{ is a min cut}]$

$= 1 - \Pr[C_1 \text{ not a min cut}] \cdot \Pr[C_2 \text{ not a min cut}]$

$p(n) \geq 1 - \left(1 - \frac{1}{2}p\left(\frac{n}{\sqrt{2}}\right)\right)\left(1 - \frac{1}{2}p\left(\frac{n}{\sqrt{2}}\right)\right) = 1 - \left(1 - \frac{1}{2}p\left(\frac{n}{\sqrt{2}}\right)\right)^2$

$= p\left(\frac{n}{\sqrt{2}}\right) - \frac{1}{4}p\left(\frac{n}{\sqrt{2}}\right)^2 \quad (\bigstar)$

Sketch: $f(t) = p(\sqrt{2}^t)$

(assuming $n = \sqrt{2}^t$)

$(\bigstar)$ becomes

$f(t) \geq f(t-1) - \frac{1}{4}f(t-1)^2$ "solves to" $p(n) = \Omega\left(\frac{1}{\log n}\right)$

Improved algorithm: Karger-Stein

Theorem. [Karger-Stein 1996] The Karger-Stein algorithm runs in time $O(n^2 \log n)$ and returns a min cut with probability at least $\Omega(1/\log n)$.

Corollary. The "best-of-T" Karger-Stein algorithm runs in time $O(n^2 \log^2 n)$ and returns a min cut with probability at least 99%.

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Best known. [Karger 2000] $O(m \log^3 n)$.

And now, for something completely different

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Theorem. An undirected graph G=(V,E) has at most _____ distinct min cuts.

And now, for something completely different?

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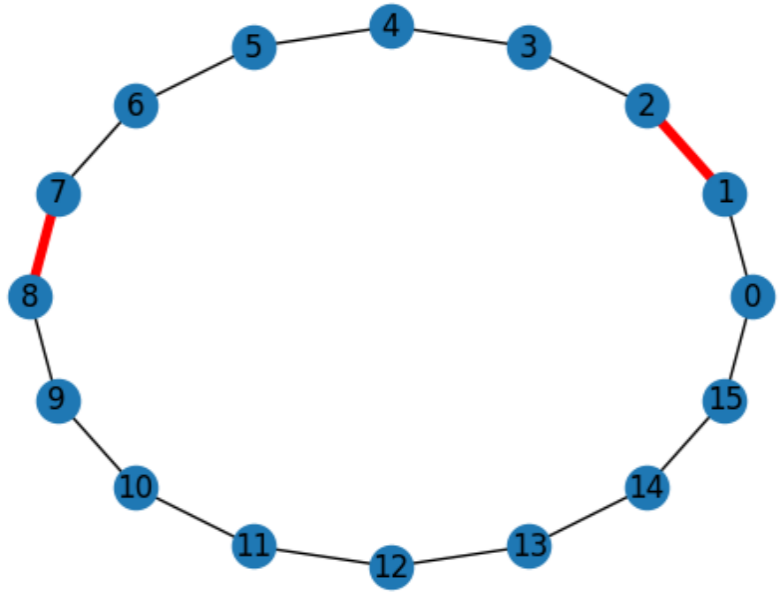
And now, for something completely different?

Theorem. An undirected graph G=(V,E) has at most n(n-1)/2 distinct min cuts. And this is tight.



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Proof. Karger's algo returns any fixed min cut C w/proba $\geq \frac{1}{\binom{n}{2}}$.

If there are M distinct min-cuts, it'll return one of them w/proba $\geq \frac{M}{\binom{n}{2}}$

$$1 \geq \Pr[\text{Karger's algo succeeds}] \geq \frac{M}{\binom{n}{2}}.$$

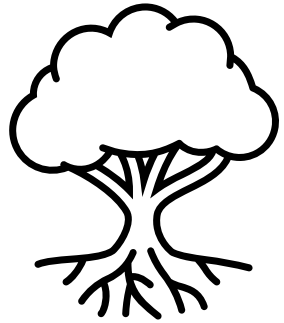
so $M \leq \binom{n}{2}$. □

And now, for something completely different?

How mind-blowing is this? Analyzing an algorithm proved a seemingly unrelated mathematical statement. 🤯

Compare this with last time, and the probabilistic method ("proof without algorithm!").

And now, for something else completely different!



Minimum Spanning Tree in (Expected) Linear Time: Karger-Klein-Tarjan.

MST via KKT: expected $O(m)$ time. *

Don't know any $O(m)$ time det. algo!

Kruskal: $O(m \log n)$
Bum's: $O(m + n \log n)$
Chazelle: $O(m \cdot \alpha(n))$
↑ some very slow-growing function

* You don't need to know the proof, but you need to know the statement and the key idea(s) of the algo] see lecture notes